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Enhancing Zero-Shot Relation Extraction through Staged Interaction with Large Language Models

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***Abstract*—Zero-shot Relation Triplet Extraction (ZeroRTE) is a challenging yet valuable task that extracts relation triplets from unstructured texts for new relation types, significantly reducing the time and effort needed for data labeling. With the advancement of the zero-shot capabilities of large language models, the performance of many zero-shot tasks has been further improved only simply by interacting with large language models (LLMs). In this work, we transform the zero-shot triplet extraction task into a two-stage chat with LLMs. Specifically, in the first stage, we prompt the LLMs to perform Named Entity Recognition (NER). In the second stage, we prompt the LLMs to perform Relation Classification (RC) using the results from the first stage. Experiments on Wiki-ZSL and FewRel datasets show the efficacy of Relation Prompt for the ZeroRTE task. Notably, our method significantly outperforms strong baselines, achieving an impressive 15.89% increase in F1 scores, particularly on Wiki-ZSL with 15 unseen relations.**

***Index Terms*—Relation Triplet Extraction, LLMs, Zero-Shot, Prompt, Instruction-tuning.**

I. Introduction

Relation Triplet Extraction(RTE) aims to extract a full triplet, a complete triplet, namely (head_entity, relation_type, tail_entity), from unstructured text. This process has applications in areas such as knowledge graph construction and question answering [1]. Recently, GraphRAG [2] has captured the spotlight for its innovative approach and potential. GraphRAG leverages structural information across entities to enable more precise and comprehensive retrieval and facilitating more accurate responses. The construction of the knowledge graph forms the foundation of GraphRAG, where the quality of the relation triplet extraction directly impacts its performance [3]. However, existing RTE approaches often require large datasets of annotated samples, which are costly to create, and are limited by a fixed set of relations [4]. Hence, many researchers have recognized the value of information extraction (IE) techniques with zero/few-shot methods [5] [6] [7]. Under the zero-shot setting, the relation sets used during training and testing are disjoint, as shown in the Figure 1. ZeroRTE models are trained on samples with a limited number of seen relation types and are expected to generalize to extract triplets with new or unseen relation types.

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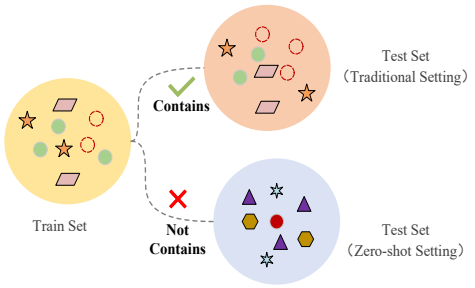


Fig. 1. Zero-shot relation triplet extraction.

With the widespread application of large language models, their zero-shot capabilities have garnered increasing attention [8]. Using carefully designed prompts has become a popular method to unlock the potential of these models for zero-shot tasks [9] [10] [11] [12]. Chain of Thought (CoT) prompting [13], which involves guiding the LLMs through a step-by-step breakdown of complex problems, has proven effective in addressing intricate issues. Although it is non-trivial to tackle ZeroRTE by decomposing it into two sub-tasks, the challenges of **How to plan the two-stage tasks?** and **How to design the prompt to enhance the ZeroRTE ability of LLMs?** still need to be addressed. RSED [14] and ChatIE [15] have decomposed ZeroRTE into Relation Selection(RS) and Entity boundary Detection, which may increase the propagation of errors, as illustrated in the Figure 2. This is because Relation Selection, the first stage, requires a deeper understanding of semantics compared to NER.

In our work, ZeroRTE is divided into two straightforward subtasks: NER and RC. By decomposing the complex ZeroRTE task into two more manageable subtasks, as opposed to Relation Selection and Entity Boundary Detection, error propagation is alleviated, which makes it easier for the model to solve the RTE. For each stage, we consider designing the prompts from three aspects: 1)Task Description, 2)Background knowledge, 3)Output Format.

Following the method described above, extensive experiments were conducted on the ZeroRTE tasks. Specifically, our approach advances the state-of-the-art RSED [14] model on two ZeroRTE datasets, achieving an improvement of up to 15.89% in the F1 score compared to the previous best model

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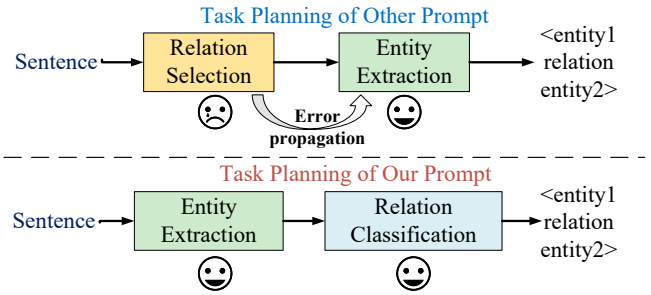


Fig. 2. Task Planning of different prompt

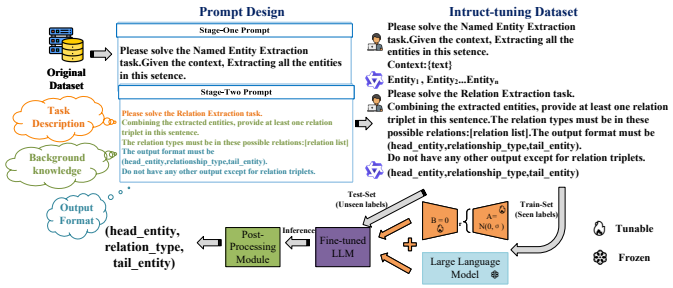


Fig. 3. Overall Architecture of Models.

on Wiki-ZSL.

The main contributions of this paper are summarized as follows:

(1) We introduced an innovative two-stage framework that leverages large language models (LLMs) for zero-shot relation triplet extraction. Our approach effectively combines NER and RC tasks, enhanced by the carefully designed prompt, which significantly reduces error propagation.

(2) We compared the impact of different two-stage task planning strategies and various prompt designs on the ZeroRTE task, providing a detailed analysis of their effects.

(3) Experiments on two datasets demonstrate that our proposed method is state-of-the-art (SOTA) in the field of ZeroRTE.

II. Method

A. Task Formulation

Let $D = \{S, < head_entity, R, tail_entity >\}$denotes the whole dataset, consisting of the input sentences S, the output triplets $< head_entity, R, tail_entity >$ where R is the set of relation labels. $D = D_s \cup D_u$, where D_s, D_u refer to the seen and unseen datasets respectively. The model is trained on D_s and evaluated on D_u. $R = R_s \cup R_u$ is predefined, comprising the seen relation label set $R_s = \{r_1^s, \cdots, r_n^s\}$ and unseen relation label set $R_u = \{r_1^u, \cdots, r_m^u\}$, where $n = |R_s|$ and $m = |R_u|$ are the number of seen and unseen relation labels respectively. $R_s \cap R_u = \emptyset$, R_s and R_u are disjoint. One sentence $s \in S$ contains one or more triplets.

B. Overview

To tackle the ZeroRTE task, we propose a two-stage dialogue framework based on the LLMs. The framework divides the task into two phases: NER and RC, with carefully designed prompts for each stage. Using these prompts, we created an instruct-tuning dataset. We then fine-tuned the LLM using the Low-Rank Adaptation (LoRA) method [16] based on this instruct-tuning dataset.

C. Two-stage Prompts Design

We have designed a two-stage prompts for zero-shot tasks, aimed at decomposing the complex relation triplet extraction problem into two more manageable subtasks. This approach is intended to improve the performance of large models in relation extraction tasks. The overall architecture of the prompt design is detailed in Figure 3.

1) Stage-One: Named Entity Recognition: ChatIE [15] and RSED [14] transform Relation Triplet Extraction into relation category selection and relation generation tasks. Our experiments reveal that large language models (LLMs) are prone to significant biases during the relation type selection phase when using this approach, which adversely affects the final performance of relation extraction. This issue arises because selecting relation types requires a nuanced understanding of semantics to make accurate choices. In contrast, entity recognition is a relatively straightforward task that involves identifying relevant entities in the text.

Consequently, in our research, we have replaced the relation type selection task with entity recognition and designed the prompts as illustrated in Figure 3.

2) Stage-Two: Relation Classification and Triplets Generation: This stage is the core phase of relation triplet extraction, responsible for combining the results of the previous entity extraction round, performing relation classification, and generating triplets that conform to the specified format. The design of prompt in this stage is mainly considered from three aspects:

(1) Task Description: In this stage, the prompt is designed to clearly specify the task of relation triplet extraction. The prompt should guide the model to generate triples that adhere to the specified format.

(2) Background Information: Instructions should be provided on how to combine the entities extracted in the previous phase with the provided context. The prompt should direct the model to perform relationship classification based on this information and to generate relation triplets in the specified format. Each triplet should consist of a subject entity, a relationship label, and an object entity.

(3) Output Format: The prompt must clearly define the expected output format for the generated relation triplets. It should specify the structure and organization of each triplet, including the order of entities and the representation of the relationship label. This ensures that the model produces output that conforms to the desired format.

Following the three design principles mentioned above, we have designed the prompt for the second stage as illustrated in Figure 3.

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D. LLMs Fine-tuning

After finalizing the prompt design, we integrated it with the dataset to create the instruct-tuning dataset for our work. Subsequently, we fine-tuned the LLMs using LoRA based on this instruction-tuning dataset.

Our model involves two training tasks: named entity recognition and relation triplets generation. We train the model in a multi-task manner. For named entity recognition, the training objective is to minimize the cross entropy loss:

$$\mathcal{L}_{ent} = -\sum_{c=1}^{C} y_{o,c} \log(p_{o,c}) \tag{1}$$

For relation triplets generation, we also adopt the cross entropy loss, and the entity loss is defined as:

$$\mathcal{L}_{rel} = -\sum_{t=1}^{T} y_{o,t} \log(p_{o,t}) \tag{2}$$

Here, $\mathcal{L}_{ent}$ denotes the loss function of the NER task. $\mathcal{L}_{rel}$ denotes the loss function of the relation triplets generation task. C represents the number of tokens for the entity to be predicted. T represents the number of tokens for the triplets to be predicted. In the NER phase, $y_{o,c}$ and $p_{o,c}$ represent the ground truth values and the predictions of the LLMs, respectively. In the relation triplets generation phase, $y_{o,t}$ and $p_{o,t}$ represent the ground truth values and the predictions of the LLMs, respectively.

We treat each loss equally and the model learns to minimize $\mathcal{L} = \mathcal{L}_{rel} + \mathcal{L}_{ent}$ jointly.

TABLE I
RESULTS COMPARED WITH BASELINE MODELS
(**BOLD** VALUES INDICATE THE BEST MODEL, WHILE UNDERLINE REPRESENTS THE SECOND-BEST.)

Unseen Labels	Method	Wiki-ZSL			Fewrel		
		P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)
m=15	RelationPrompt [17]	26.19	32.12	28.85	17.73	23.20	20.08
	PAED [18]	-	-	-	20.68	23.39	21.95
	ZETT [19]	-	-	21.21(±0.35)	-	-	24.39(±0.37)
	RSED [14]	25.37	33.80	28.98	27.00	23.55	25.16
	Ours(Llama)	56.47	36.45	44.30	**31.28**	24.87	**27.71**
	Ours(Qwen)	**57.31**	**36.87**	**44.87**	30.28	**25.03**	27.41
m=10	RelationPrompt [17]	30.20	32.31	31.19	21.59	28.68	24.61
	PAED [18]	-	-	-	23.31	27.42	25.15
	ZETT [19]	-	-	24.87(±0.32)	-	-	31.28(±0.54)
	RSED [14]	27.09	**39.09**	32.00	30.89	29.90	30.39
	Ours(Llama)	56.79	36.76	44.63	34.98	**32.45**	33.67
	Ours(Qwen)	**60.77**	37.76	**46.57**	**35.46**	32.17	**33.74**
m=5	RelationPrompt [17]	29.11	31.00	30.01	20.80	24.32	22.34
	PAED [18]	-	-	-	25.79	34.54	29.47
	ZETT [19]	-	-	31.17 (±0.69)	-	-	33.71 (±0.42)
	RSED [14]	38.14	**36.84**	**37.48**	43.91	34.97	38.93
	Ours(Llama)	42.47	20.27	27.40	50.77	**52.86**	51.79
	Ours(Qwen)	**42.52**	23.27	30.09	**53.10**	52.66	**52.88**

III. EXPERIMENTS

A. Experimental Settings

For the task of zero-shot relation triplets extraction, we fine-tune the Qwen1.5-14B-Chat [20] proposed by Alibaba Cloud and Llama2-13B-Chat [21] developed by Meta. We use the following two datasets for our experiments. FewRel [22] is hand-annotated for few-shot relation extraction, but we made it suitable for the zero-shot setting after data splitting into disjoint relation label sets for training and testing. Wiki-ZSL [23] is constructed through distant supervision over Wikipedia articles and the Wikidata knowledge base. For each dataset, we varied the number of unseen labels, setting it to $m \in \{5, 10, 15\}$, while the remaining labels were treated as seen during training in our experiments.

To evaluate multiple triplet extraction, we use the Micro F1 metric which is standard in structured prediction tasks, and report the precision (P.) and recall (R.). The experimental results represent the mean values obtained from five independent replicates.

B. Experimental Results

To demonstrate the effectiveness of our proposed method in the ZeroRTE task, we compare it with several existing ZeroRTE models.

• **RelationPrompt** [17] fine-tuned the BART [24] on the synthetic relation sentences generated by prompting language models.

• **PAED** [18] presented a generation-based framework for zero-shot PAED. A novel HNS strategy and a Meta-VAE sampler with CSC are presented to enhance the performance of this model.

• **ZETT** [19] resolved the Zero-shot Triplet extraction by Template infilling based on an end-to-end generative language model.

• **RSED** [14] proposed a method with potential candidate relation selection and entity boundary detection directly utilizing the semantics of unseen relations to tackle the ZeroRTE task.

The experimental results are reported in Table I. Our framework consistently outperforms existing methods. For instance, with 15 unseen labels, our framework improves the F1 score by 2.25% on the Fewrel dataset and by 15.89% on the Wiki-ZSL dataset, compared to the RSED, which is the previous state-of-the-art method. In conclusion, our proposed method demonstrates a superior ability to leverage the potential of LLMs in the RTE task.

However, it is important to note that when the number of unseen labels is 5, our method's performance on the Wiki-ZSL dataset is not as strong as that of the RSED method. This discrepancy may be attributable to the distribution characteristics of the Wiki-ZSL dataset. Additionally, our observations suggest that with fewer labels, there is a higher likelihood of generating triplets that fall outside the predefined relations, which could contribute to the observed performance decline. Despite this, our method still outperforms other models. Addressing this issue will be a key focus of our future work.

IV. ANALYSIS

In this section, we focus on two main research questions proposed in the Introduction Section. The experiments in this section are all conducted using the Qwen model and verified on the Fewrel dataset.

A. How to plan the two-stage tasks?

To compare the impact of different two-stage task settings on model performance, we designed other two-stage models

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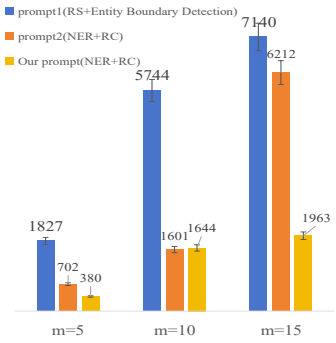


Fig. 4. Number of Predictions Outside Pre-defined List.

with different content for the experiments. Taking the FewRel as the test dataset, we conducted experiments under a different number of unseen label settings, and the experimental results are shown in Table II.

• **prompt1(RS+Entity boundary Detection)**

This prompt still employs a two-stage task framework. However, unlike the prompt used in this paper, we have modified the first stage to perform RS instead of the NER task.

TABLE II
RESULTS WITH DIFFERENT PROMPT

Unseen Labels	Method	P(%)	R(%)	F1(%)
m=15	*w.t. prompt1*	**37.20**	11.91	18.04
	w.t. prompt2	17.73	23.20	20.08
	Our prompt	30.28	**25.03**	**27.41**
m=10	*w.t. prompt1*	**51.59**	9.26	15.70
	w.t. prompt2	35.58	28.24	31.49
	Our prompt	35.46	**32.17**	**33.74**
m=5	*w.t. prompt1*	**67.72**	32.37	43.80
	w.t. prompt2	50.85	41.09	45.45
	Our prompt	53.10	**52.66**	**52.88**

The results indicate that using **prompt1**, which involves selecting relation types first and then generating triplets, performs poorly. As illustrated in Figure 4, the number of predictions outside the predefined relation list is higher with **prompt1**. This observation underscores the critical importance of task selection in a two-stage approach for relation triplet extraction.

The poor performance observed with the Prompt1 task setup can be attributed to its initial phase of relation selection, which demands a deep semantic understanding. This increased complexity heightens the risk of errors being propagated, leading to a higher incidence of incorrect relationship types or instances that fall outside the predefined list.

Consequently, we conclude that the two-stage task setup involving NER followed by RC is more effective in minimizing predictions outside the predefined relation list, thereby improving performance on the Zero-RTE task.

B. How to design the prompt to enhance the ZeroRTE ability of LLMs?

• **prompt2(NER+RC)** This prompt adopts the same two-stage task setup as this paper, but its task description is cumbersome.

From Figure 4 and Table II, we observed that **prompt2** shows slightly inferior performance compared to the prompt used in this paper. Subsequently, we conducted ablation experiments on each component of our designed prompts. As shown in Table III, the removal of any component leads to a decline in performance.

Therefore, we can conclude that a more concise and clear prompt is more conducive to the LLM's understanding and completion of downstream tasks.

TABLE III
DIFFERENT DESIGN OF THE PROMPT

Unseen Labels	Method	P(%)	R(%)	F1(%)
m=15	*w.o.Background Information*	**37.21**	21.46	27.22
	w.o.Output Format	25.78	5.59	9.18
	Ours	30.28	**25.03**	**27.41**
m=10	*w.o.Background Information*	**42.18**	25.65	31.90
	w.o.Output Format	33.52	10.37	15.84
	Ours	35.46	**32.17**	**33.74**
m=5	*w.o.Background Information*	52.80	28.91	37.36
	w.o.Output Format	51.70	16.12	24.58
	Ours	**53.10**	**52.66**	**52.88**

C. Other Ablation Study

TABLE IV
ABLATION STUDY RESULTS

Unseen Labels	Method	P(%)	R(%)	F1(%)
m=15	*w.o.first-stage prompt*	29.42	24.69	26.84
	w.o.LoRA	22.06	8.51	12.29
	Ours	**30.28**	**25.03**	**27.41**
m=10	*w.o.first-stage prompt*	33.30	26.93	29.78
	w.o.LoRA	17.18	4.76	7.45
	Ours	**35.46**	**32.17**	**33.74**
m=5	*w.o.first-stage prompt*	51.29	40.49	45.25
	w.o.LoRA	43.59	11.09	17.68
	Ours	**53.10**	**52.66**	**52.88**

We first removed the prompt from the first stage, using only the second stage's prompt and input text. Subsequently, we eliminated the fine-tuning module and relied solely on the designed prompts for model inference.

The results show a substantial decline in model performance, indicating that both the two-stage prompts approach and LoRA fine-tuning are crucial for enhancing the performance of LLMs in the ZeroRTE task.

V. CONCLUSIONS

This paper addresses the Zero-shot Relation Triplet Extraction (ZeroRTE) task and introduces an innovative two-stage conversational approach to improve the extraction of relation triplets from unstructured texts involving previously unseen relation types. The experimental results demonstrate that our method significantly outperforms established baselines.

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